

Agentic AI for Banking

From Documents to Decisions — Autonomously

SFI Masterclass

2026

Part I — Concepts

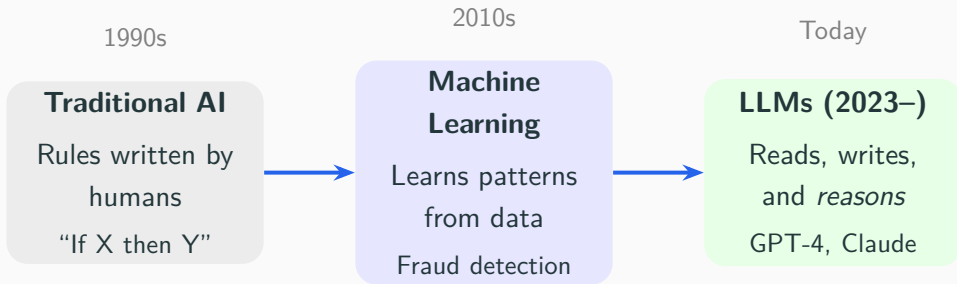
1. The AI Revolution in 60 Seconds
2. What LLMs Can (and Can't) Do
3. RAG: Grounding AI in Facts
4. Agents: AI That *Acts*

Part II — Live Demo

5. Hands-On: Analysing 10-K Filings
6. From Prototype to Production
7. The Banking Opportunity
8. Risks & Governance

The AI Revolution

What Changed?



The breakthrough

For the first time, machines can **read**, **understand**, and **generate** human-quality text — including financial documents.

What LLMs Can (and Can't) Do

LLMs: Strengths & Limitations

✓ Can Do

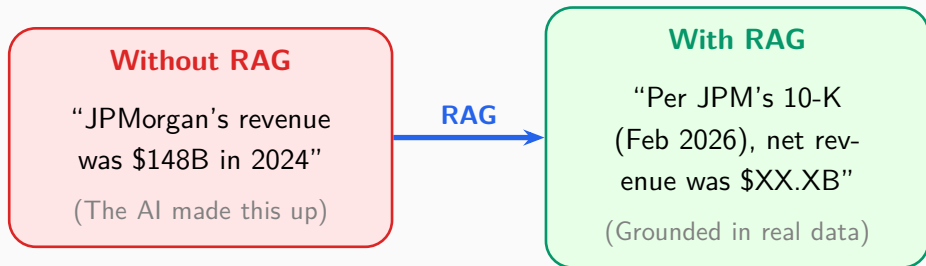
- Summarise 200-page filings in seconds
- Compare documents side by side
- Extract structured data from text
- Draft memos, emails, reports
- Reason across multiple sources

✗ Cannot Do (Alone)

- Access your internal data
- Guarantee factual accuracy
- Perform multi-step research
- Use your tools & databases
- Explain its confidence level

Solution: RAG + Agents fill these gaps.

The Hallucination Problem

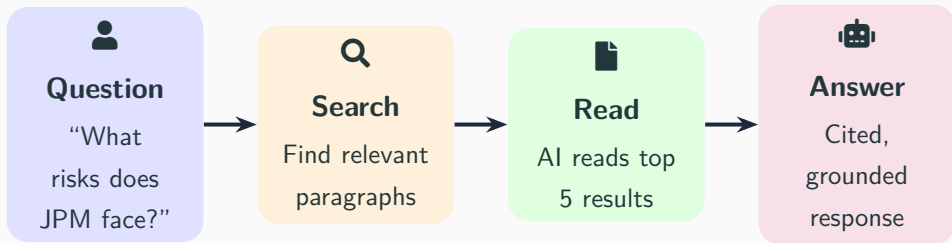


In banking, **accuracy is non-negotiable**. RAG ensures every AI answer is traceable to a source document.

RAG: Grounding AI in Facts

What is RAG?

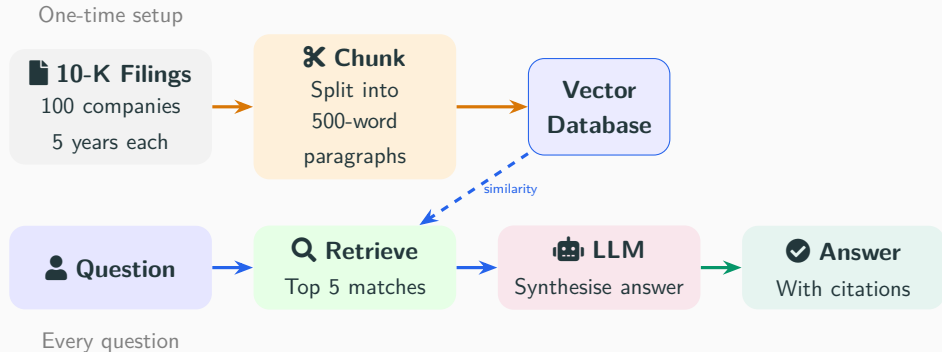
Retrieval-Augmented Generation = Search + AI



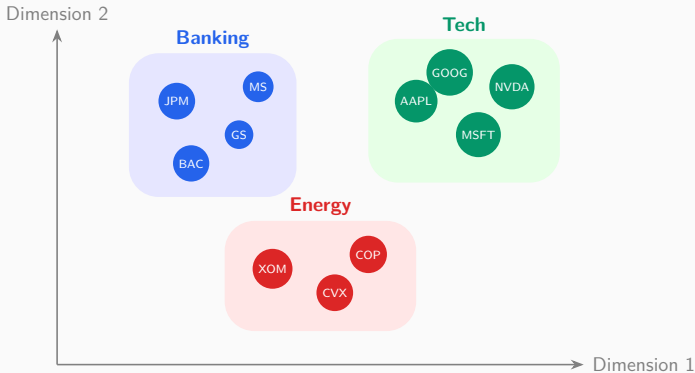
The Analyst Analogy

RAG works like a junior analyst: **search** the filing cabinet, **read** the relevant pages, **write** a summary with citations.

How RAG Works Under the Hood



Embeddings: How Machines “Understand” Text



Companies that use **similar language** end up **close together** in vector space.
A query about “credit risk” will land near the banking cluster — not near energy or tech.

Agents: AI That Acts

From Chatbot to Agent

Chatbot

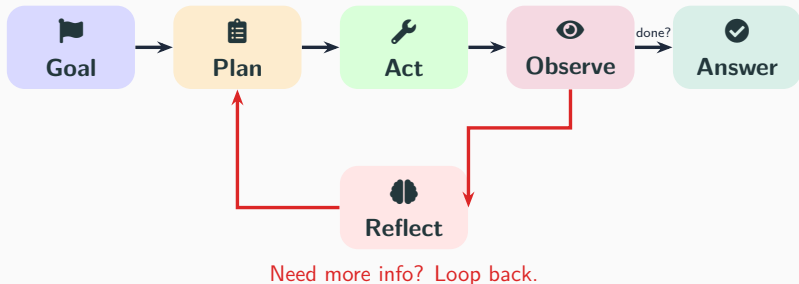
- › Single question → single answer
- › No memory between turns
- › Cannot use tools
- › Cannot plan



Agent

- › Complex goal → multi-step plan
- › Remembers what it found
- › Uses tools (search, calculate)
- › Reflects and adapts

The Agentic Loop



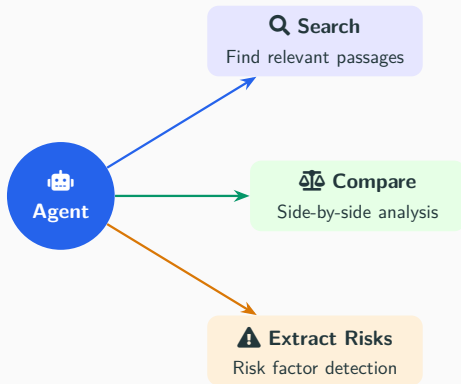
Example

Goal: "Compare credit risk at JPM vs GS"

Plan: Search JPM risks → Search GS risks → Compare → Synthesise

Reflect: "I found loan loss provisions but not derivatives exposure — let me search again."

Agent Tools = Analyst Toolkit



Like a Bloomberg terminal for an analyst — the agent decides **which tool** to use and **when**, based on the question.

Hands-On: Analysing 10-K Filings

Our Dataset

458 SEC 10-K Filings

100 S&P 100 companies $\times$ up to 5 years

Tech

Banking

Health

Energy

Consumer

Industrial

- All filings publicly hosted on GitHub — **no API keys needed**
- Accessible from Google Colab via HTTPS
- Real SEC data, not synthetic

The Pipeline We Built



Let's switch to the live notebook.



Live Demo

Part I: TF-IDF + Rule-Based Agent

Sections 1–8 of the notebook

From Prototype to Production

The Upgrade Path

Level 0 Ctrl+F in PDF files

Manual, doesn't scale



Level 1 TF-IDF + FAISS vector search Free, fast, keyword matching



Level 2 OpenAI embeddings + RAG
understanding

Semantic



Level 3 LLM agent with tool use Autonomous multi-step reasoning



Level 4 Multi-agent systems

Specialist agents collaborate

Today's demo

With OpenAI API

TF-IDF vs OpenAI Embeddings

	TF-IDF	OpenAI Embeddings
Approach	Count words, weight by rarity	Neural network trained on billions of texts
Synonyms?	“revenue” $\neq$ “sales”	“revenue” $\approx$ “sales”
Context?	“bank” (river) = “bank” (finance)	Understands the difference
Cost	Free	\$0.02 per 1M tokens
Best for	Prototyping, keyword search	Production, semantic search

In financial text, this matters: “provisions for credit losses”, “allowance for loan losses”, and “expected credit losses” all mean the same thing. Only neural embeddings understand this.



Live Demo

Part II: OpenAI Embeddings + LLM Agent

Sections 10–13 of the notebook

The Banking Opportunity

Use Cases Across the Bank



Risk Mgmt

“Flag material litigation across 50 counterparty filings”



Credit Analysis

“Compare debt covenants across 10 borrowers”



Compliance

“Do portfolio companies disclose TCFD climate risks?”



M&A Due Diligence

“Review target’s risk factors and contingent liabilities”



Audit

“Verify extracted figures match the original filing”

Impact: Time Savings



Key insight

The value isn't replacing analysts — it's **freeing them** for higher-order judgment.

Risks & Governance

What Can Go Wrong

Technical Risks

- **Hallucination** — AI states facts not in the source
- **Retrieval failure** — wrong documents fetched
- **Stale data** — filings not updated
- **Context limits** — can't read entire filings at once

Organisational Risks

- **Over-reliance** — trusting AI without verification
- **Data leakage** — sensitive data sent to APIs
- **Regulatory** — explainability requirements
- **Liability** — who's responsible for AI errors?

Governance Framework

Human-in-the-Loop — Every AI output reviewed before action

Source Citations — Every claim traceable to a filing & page

Audit Trail — Full log of reasoning, tool calls, and sources

Access Controls — Data classification, API keys, PII filtering

Defence in
depth

Key Takeaways

1. **RAG eliminates hallucination** by grounding every answer in real documents
2. **Agents go beyond Q&A** — they plan, search, reflect, and iterate
3. **The technology is ready today** — OpenAI embeddings + GPT-4o + FAISS
4. **Start small, scale fast** — prototype in a notebook, deploy behind your firewall
5. **Governance is essential** — human oversight, citations, and audit trails

Thank You

Questions?



`github.com/SMalamud/sec-10k-downloader`



Open in Colab: click the link in the README